Baker Charters School

Honors Algebra 2B

# Baker Web Academy School Notes

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**Mar 18 2022 Fri (23:11:52)**

**Lesson 1: Intercepts/Asymptotes/Domain/Range** **Unit 5**

**Definition 1.** (Intercept) When the line intersects/touches/goes through/etc the $x$ or $y$ axis. If the line touches the $x$-axis, then it's a $x$-intercept. Likewise, if the line touches the $y$-axis, then it's a $y$-intercept.

**Definition 2.** (Asymptotes) A line that a curve approaches as it heads toward infinity.

**Definition 3.** (Domain) The set of all numbers that appear as $x$-coordinates of points on the graph.

**Definition 4.** (Range) The set of all numbers that appear as $y$-coordinates of points on the graph.

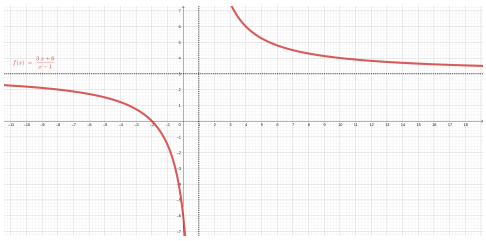


**Figure 1:** Vertical and Horizontal Asymptotes

**Find Vertical/Horizontal asymptotes**

- Vertical Asymptote(s):

  To find the vertical asymptote, you try to find where the graph of the function are approaching the $y$-axis, but never touch. From this, we conclude that the vertical asymptote is $x = 1$.

- Horizontal Asymptote(s)

  To find the horizontal asymptote, you try to find where the graph of the function are approaching the $x$-axis, but never touch. From this, we conclude that the vertical asymptote is $y = 3$.

**Domain and Range**

- Domain:

  The domain is usually the set of all real numbers except the values of the vertical asymptotes. That's only if the graph doesn't have any holes. We found the vertical asymptote as $x = 1$. So, here's the domain: $(-\infty, 1) \cup (1, \infty)$

- Range:

  We project the graph onto the $y$-axis. The numbers that appear as $y$-values of points on the graph will make our range. Remember not to include the horizontal asymptote. So, here's our range: $(-\infty, 3) \cup (3, \infty)$

**Find all $x$-intercepts and $y$-intercepts**

- Find all $x$-intercepts:

To find the $x$-intercepts, just look for when the graph passes through the $x$-axis. So, the $x$-intercept is $-2$.

- Find all $y$-intercepts:

  To find the $y$-intercepts, just look for when the graph passes through the $y$-axis. So, the $y$-intercept is $-6$.

# List of Theorems

## List of Postulates

# List of Conjectures

# List of Corollaries

# List of Definitions

1 Definition (Intercept) . . . . . Page: 1
2 Definition (Asymptotes) . . . . . Page: 1
3 Definition (Domain) . . . . . Page: 1
4 Definition (Range) . . . . . Page: 1

# List of Reviews

# List of Properties

# List of Propositions

# Essential Formulas for Algebra 2 Final Exam

## Laws of Exponents

| | |
|---|---|
| Multiply Powers of the Same Base = Adding Exponents | $(a^m)(a^n) = a^{m+n}$ |
| Divide Powers of the Same Base = Subtracting Exponents | $\frac{a^m}{a^n} = a^{m-n}$ |
| Power Rule = Multiplying Exponents | $(a^m)^n = a^{m \times n}$ |
| Zero Exponent = 1 | $a^0 = 1$ |
| Distribution of Exponent with Multiple Bases | $(ab)^n = a^n b^n$<br>$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$ |
| Negative Exponent = Reciprocal | $a^{-n} = \frac{1}{a^n}$<br>$\frac{a^{-m}}{b^{-n}} = \frac{b^n}{a^m}$ |
| Distribution of Negative Exponent with Multiple Bases | $(ab)^{-n} = a^{-n}b^{-n} = \frac{1}{a^n b^n}$<br>$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n = \frac{b^n}{a^n}$ |

$$\sqrt{a+b} \neq \sqrt{a} + \sqrt{b} \qquad \sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$$

$$\sqrt{a-b} \neq \sqrt{a} - \sqrt{b} \qquad \sqrt{a \div b} = \sqrt{a} \div \sqrt{b}$$

## Properties of Radicals

| | |
|---|---|
| Distribution of Radicals of the Same Index<br>(where $a \geq 0$ and $b \geq 0$ if $n$ is even) | $\sqrt[n]{ab} = \left(\sqrt[n]{a}\right)\left(\sqrt[n]{b}\right)$<br>$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$ |
| Power Rule of Radicals = Multiplying Exponents | $\sqrt[m]{\sqrt[n]{a}} = \sqrt[(m \times n)]{a}$ |
| Reverse Operations of Radicals and Exponents | $\sqrt[n]{a^n} = a$ (if $n$ is odd)<br>$\sqrt[n]{a^n} = \lvert a \rvert$ (if $n$ is even) |

$$a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

**The index of the radical is the denominator of the fractional exponent.**

**Special Products**

$(A + B)^2 = A^2 + 2AB + B^2$

$(A - B)^2 = A^2 - 2AB + B^2$

$(A + B)(A - B) = A^2 - B^2$

$(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$

$(A - B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$

**Special Expressions**

| | |
|---|---|
| Difference of Squares | $A^2 - B^2 = (A + B)(A - B)$ |
| Perfect Trinomial Squares | $A^2 + 2AB + B^2 = (A + B)^2$ |
| Perfect Trinomial Squares | $A^2 - 2AB + B^2 = (A - B)^2$ |
| Sum of Cubes | $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$ |
| Difference of Cubes | $A^3 - B^3 = (A - B)(A^2 + AB + B^2)$ |

**Quadratic Formula:** $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

**Discriminant** $= b^2 - 4ac$

When Discriminant is *Positive*, $b^2 - 4ac > 0 \rightarrow$ *Two Distinct Real Roots*

When Discriminant is *Zero*, $b^2 - 4ac = 0 \rightarrow$ *One Distinct Real Root (or Two Equal Real Roots)*

When Discriminant is *Negative*, $b^2 - 4ac < 0 \rightarrow$ *No Real Roots*

***Note the pattern:***

$i^1 = i \quad i^2 = -1 \quad i^3 = -i \quad i^4 = 1$

$i^5 = i \quad i^6 = -1 \quad i^7 = -i \quad i^8 = 1$

$i^9 = i \quad i^{10} = -1 \quad \ldots..$

Pattern repeats every $4^{th}$ power of $i$.

**Product of Conjugate Complex Numbers**

$(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 - b^2(-1)$

$(a + bi)(a - bi) = a^2 + b^2$

**Midpoint of a Line Segment**

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

**Distance of a Line Segement**

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

**Slope**

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

**Standard Equation for Circles**

$$(x - h)^2 + (y - k)^2 = r^2$$

$P(x, y)$ = any point on the path of the circle

$C(h, k)$ = centre of the circle

$r$ = length of the radius

**Point-Slope form**: - a form of a linear equation when given a slope ($m$) and a point ($x_1$, $y_1$) on the line

$$\frac{y - y_1}{x - x_1} = m \text{ (slope formula)} \qquad y - y_1 = m(x - x_1) \quad \text{(Point-Slope form)}$$

If we rearrange the equations so that all terms are on one side, it will be in **standard (general) form**:

$$Ax + By + C = 0 \qquad \text{(Standard or General form)}$$

($A \geq 0$, the leading coefficient for the $x$ term must be positive)

When given a slope ($m$) and the $y$-intercept (0, $b$) of the line, we can find the equation of the line using the **slope and $y$-intercept form**:

$$y = mx + b$$

where $m$ = slope and $b$ = $y$-intercept

| Parallel Lines | Perpendicular Lines |
|---|---|
| slope of line 1 = slope of line 2 | slope of line 1 = negative reciprocal slope of line 2 |
| $m_1 = m_2$ | $m_{l_1} = \frac{-1}{m_{l_2}}$ |

$y \propto x$ ($y$ is directly proportional to $x$)

$$y = kx$$

where $k$ = constant of variation (constant of proportionality – rate of change)

$y \propto \frac{xz}{w}$ ($y$ is jointly proportional to $x$, $z$ and $w$)

$$y = k\frac{xz}{w}$$

where $k$ = constant of variation (constant of proportionality)

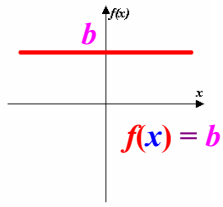
Average Rate of Change = $m = \frac{\Delta y}{\Delta x}$

Average Rate of Change = $\frac{f(b) - f(a)}{b - a}$

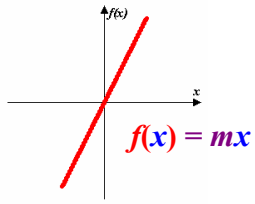
It is the slope of the secant line between the points ($a$, $f(a)$) and ($b$, $f(b)$)

## Summary of Types of Functions: (see page 226 of textbook)

### Linear Functions $f(x) = mx + b$




**Domain:** $x \in R$
**Range:** $f(x) \in R$




**Domain:** $x \in R$
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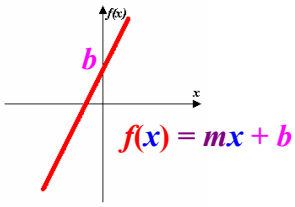
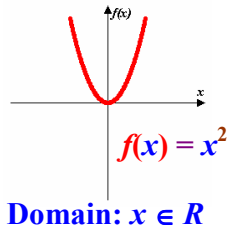
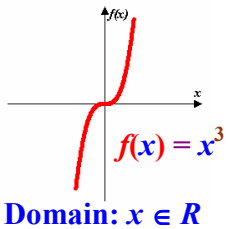



**Domain:** $x \in R$
**Range:** $f(x) \in R$

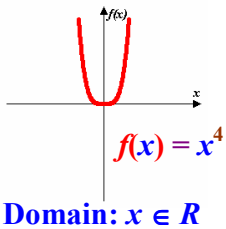
### Power Functions $f(x) = x^n$ where $n > 1$ and $n \in N$



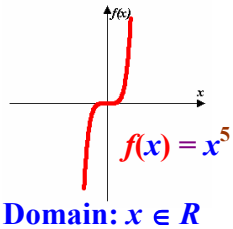

**Domain:** $x \in R$
**Range:** $f(x) \geq 0$




**Domain:** $x \in R$
**Range:** $f(x) \in R$

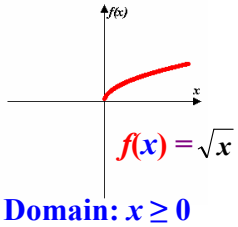



**Domain:** $x \in R$
**Range:** $f(x) \geq 0$

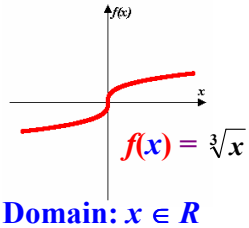



**Domain:** $x \in R$
**Range:** $f(x) \in R$

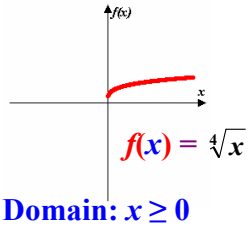
### Root Functions $f(x) = \sqrt[n]{x}$ where $n \geq 2$ and $n \in N$



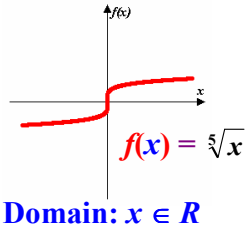

**Domain:** $x \geq 0$
**Range:** $f(x) \geq 0$




**Domain:** $x \in R$
**Range:** $f(x) \in R$

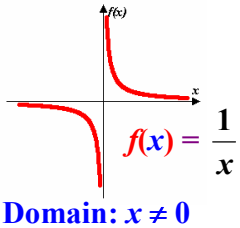



**Domain:** $x \geq 0$
**Range:** $f(x) \geq 0$

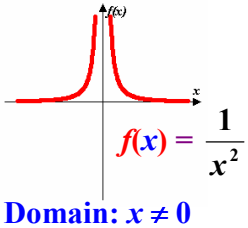



**Domain:** $x \in R$
**Range:** $f(x) \in R$

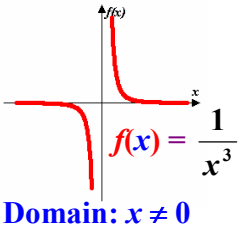
### Reciprocal Functions $f(x) = \dfrac{1}{x^n}$ where $n \in N$



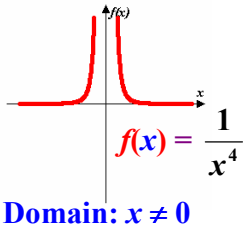

**Domain:** $x \neq 0$
**Range:** $f(x) \neq 0$




**Domain:** $x \neq 0$
**Range:** $f(x) > 0$

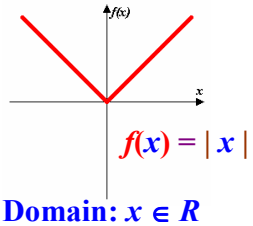



**Domain:** $x \neq 0$
**Range:** $f(x) \neq 0$



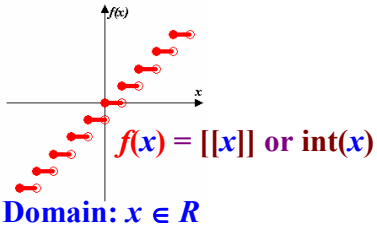

**Domain:** $x \neq 0$
**Range:** $f(x) > 0$

### Absolute Value Functions




**Domain:** $x \in R$
**Range:** $f(x) \geq 0$

### Greatest Integer Functions




**Domain:** $x \in R$
**Range:** $f(x) \in I$

$g(x) = f(x + h) + k$

$h$ = amount of horizontal movement — $h > 0$ (move left); $h < 0$ (move right)

$k$ = amount of vertical movement — $k > 0$ (move up); $k < 0$ (move down)

**Reflection off the $x$-axis**

$g(x) = -f(x)$

All values of $y$ has to switch signs but all values of $x$ remain unchanged.

**Reflection off the $y$-axis**

$g(x) = f(-x)$

All values of $x$ has to switch signs but all values of $y$ remain unchanged.

**Vertical Stretching and Shrinking**

$g(x) = af(x)$

$a$ is the Vertical Stretch Factor

$a > 1$ (Stretches Vertically by a factor of $a$)

$0 < a < 1$ (Shrinks Vertically by a factor of $a$)

**Horizontal Stretching and Shrinking**

$g(x) = f(bx)$

$b$ is the Horizontal Stretch Factor

$0 < b < 1$ (Stretches Horizontally by a factor of $1/b$)

$b > 1$ (Shrinks Horizontally by a factor of $1/b$)

**For Quadratic Functions in Standard Form of $f(x) = a(x - h)^2 + k$**

Vertex at $(h, k)$ — Axis of Symmetry at $x = h$ — Domain: $x \in R$

**$a$ = Vertical Stretch Factor**

$a > 0$ Vertex at Minimum (Parabola opens UP) — Range: $y \geq k$ (Minimum)

$a < 0$ Vertex at Maximum (Parabola opens DOWN) — Range: $y \leq k$ (Maximum)

$|a| > 1$ Stretched out Vertically — $|a| < 1$ Shrunken in Vertically

**$h$ = Horizontal Translation** (Note the standard form has $x - h$ in the bracket!)

$h > 0$ Translated Right — $h < 0$ Translated Left

**$k$ = Vertical Translation**

$k > 0$ Translated Up — $k < 0$ Translated Down

**For Quadratic Functions in General Form: $f(x) = ax^2 + bx + c$**

$y$-intercept at $(0, c)$ by letting $x = 0$ (Note: Complete the Square to change to *Standard Form*)

$x$-intercepts at $\left(\dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}, 0\right)$ if $b^2 - 4ac \geq 0$. No $x$-intercepts when $b^2 - 4ac < 0$

Vertex locates at $x = -\dfrac{b}{2a}$ $\quad y = f\left(-\dfrac{b}{2a}\right)$ Minimum when $a > 0$ ; Maximum when $a < 0$

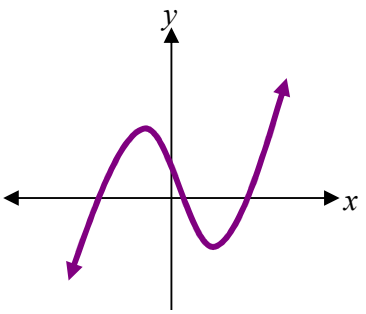
$f(x)$ = One-to-One Function $(x, y)$ — $f^{-1}(x)$ = Inverse Function $(y, x)$

Domain of $f(x) \rightarrow$ Range of $f^{-1}(x)$

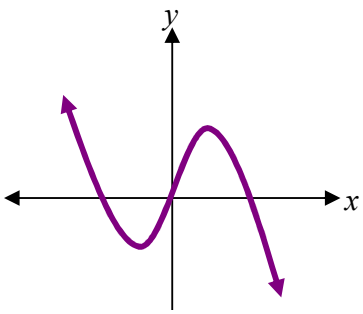
Range of $f(x) \rightarrow$ Domain of $f^{-1}(x)$

Note: $f^{-1}(x) \neq \dfrac{1}{f(x)}$ (Inverse is DIFFERENT than Reciprocal)

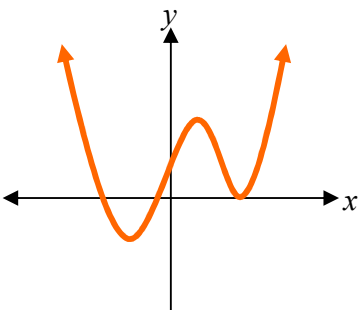
## End Behaviours and Leading Terms



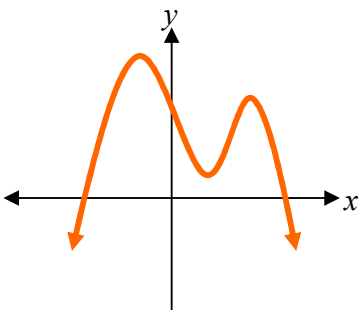

**Odd Degree Polynomial Function and Positive Leading Coefficient, $a > 0$**




**Odd Degree Polynomial Function and Negative Leading Coefficient, $a < 0$**




**Even Degree Polynomial Function and Positive Leading Coefficient, $a > 0$**




**Even Degree Polynomial Function and Negative Leading Coefficient, $a < 0$**

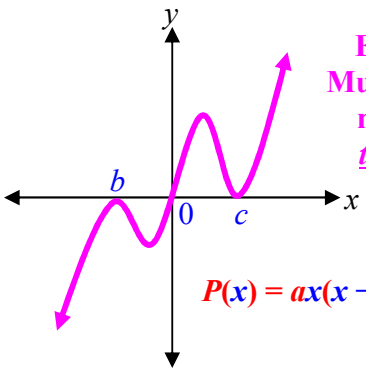
**Odd Degree Polynomial Functions**

**When $a > 0$, Left is Downward ($y \to -\infty$ as $x \to -\infty$) and Right is Upward ($y \to \infty$ as $x \to \infty$).**
**When $a < 0$, Left is Upward ($y \to \infty$ as $x \to -\infty$) and Right is Downward ($y \to -\infty$ as $x \to \infty$).**

**Even Degree Polynomial Functions**

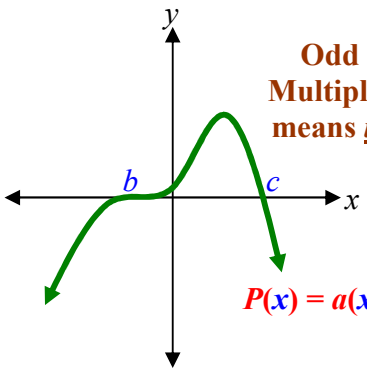
**When $a > 0$, Left is Upward ($y \to \infty$ as $x \to -\infty$) and Right is Upward ($y \to \infty$ as $x \to \infty$).**
**When $a < 0$, Left is Downward ($y \to -\infty$ as $x \to -\infty$) and Right is Downward ($y \to -\infty$ as $x \to \infty$).**

**Multiplicity**: - when a factored polynomial expression has exponents on the factor that is greater than 1.




**Even Multiplicity or Multiplicities of 2, 4, 6… means $x$-intercept is *tangent to the x-axis***

$$P(x) = ax(x-b)^{2,\,4,\,6\,\cdots}(x-c)^{2,\,4,\,6,\ldots}$$




**Odd Multiplicity or Multiplicities of 3, 5, 7… means *inflection* at $x$-int**

$$P(x) = a(x-b)^{3,\,5,\,7\,\cdots}(x-c)$$

**Polynomial Function** **Divisor Function**

**In general, for $P(x) \div D(x)$, we can write**

$$\frac{P(x)}{D(x)} = Q(x) + \frac{R}{D(x)} \quad \text{or} \quad P(x) = D(x)\,Q(x) + R$$

**Restriction: $D(x) \neq 0$**

**Quotient Function** **Remainder**

If $R = 0$ when $\frac{P(x)}{(x-b)}$, then $(x - b)$ is a factor of $P(x)$ and $P(b) = 0$.

$$P(x) = D(x) \times Q(x)$$

$P(x)$ = Original Polynomial $\quad$ $D(x)$ = Divisor (Factor) $\quad$ $Q(x)$ = Quotient

If $R \neq 0$ when $\frac{P(x)}{(x-b)}$, then $(x - b)$ is NOT a factor of $P(x)$.

$$P(x) = D(x) \times Q(x) + R(x)$$

**The Remainder Theorem**:

To find the remainder of $\frac{P(x)}{x-b}$: Substitute $b$ from the Divisor, $(x - b)$, into the Polynomial, $P(x)$.

In general, when $\frac{P(x)}{x-b}$, $P(b)$ = Remainder.

To find the remainder of $\frac{P(x)}{ax-b}$: Substitute $\left(\frac{b}{a}\right)$ from the Divisor, $(ax - b)$, into the Polynomial, $P(x)$.

In general, when $\frac{P(x)}{ax-b}$, $P\left(\frac{b}{a}\right)$ = Remainder.

**The Factor Theorem**:

1. If $\frac{P(x)}{x-b}$ gives a Remainder of 0, then $(x - b)$ is the Factor of $P(x)$.

OR

If $P(b) = 0$, then $(x - b)$ is the Factor of $P(x)$.

2. If $\frac{P(x)}{ax-b}$ gives a Remainder of 0, then $(ax - b)$ is the Factor of $P(x)$.

OR

If $P\left(\frac{b}{a}\right) = 0$, then $(ax - b)$ is the Factor of $P(x)$.

**Rational Roots Theorem**:

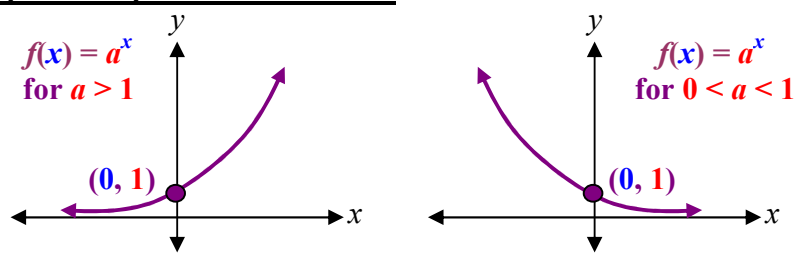
For a polynomial $P(x)$, a List of POTENTIAL Rational Roots can be generated by Dividing ALL the Factors of its Constant Term by ALL the Factors of its Leading Coefficient.

$$\text{Potential Rational Zeros of } P(x) = \frac{\text{ALL Factors of the Constant Term}}{\text{ALL Factors of the Leading Coefficient}}$$

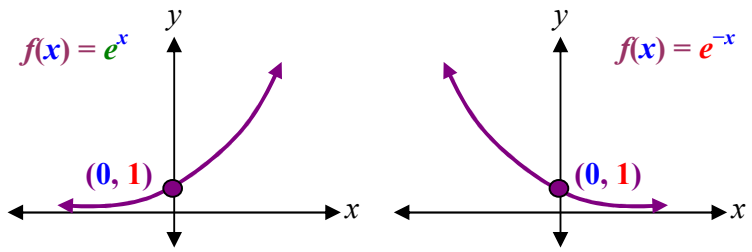
**The Zero Theorem**

There are $n$ number of solutions (complex, real or both) for any $n^{th}$ degree polynomial function accounting that that a zero with multiplicity of $k$ is counted $k$ times.

## Graphs of Exponential Functions




## Graphs of Natural Exponential Functions




$$y = a^x \longleftrightarrow x = \log_a y$$

## Simple Properties of Logarithms

| | |
|---|---|
| $\log_a 1 = 0$ | because $a^0 = 1$ |
| $\log_a a = 1$ | because $a^1 = a$ |
| $a^{\log_a x} = x$ | because exponent and logarithm are inverse of one another |
| $\log_a a^x = x$ | because logarithm and exponent are inverse of one another |

## Common and Natural Logarithm

| | | | |
|---|---|---|---|
| Common Logarithm: | $\log x = y$ | $\longleftrightarrow$ | $10^y = x$ |
| Natural Logarithm: | $\ln x = y$ | $\longleftrightarrow$ | $e^y = x$ |

| Exponential Laws | Logarithmic Laws |
|---|---|
| $(a^m)(a^n) = a^{m+n}$ | $\log_a x + \log_a y = \log_a(xy)$ |
| $\dfrac{a^m}{a^n} = a^{m-n}$ | $\log_a x - \log_a y = \log_a\left(\dfrac{x}{y}\right)$ |
| $(a^m)^n = a^{m \times n}$ | $\log_a x^y = y \log_a x$ |
| $a^0 = 1$ | $\log_a 1 = 0$ |

## Common Logarithm Mistakes

$\log_a(x+y) \neq \log_a x + \log_a y$

Example: $\log(2+8) \neq \log 2 + \log 8$

$1 \neq 0.3010 + 0.9031$

$\log_a(x-y) \neq \log_a x - \log_a y$

Example: $\log(120-20) \neq \log 120 + \log 20$

$2 \neq 2.0792 + 1.3010$

$\log_a\left(\frac{x}{y}\right) \neq \frac{\log_a x}{\log_a y}$

Example: $\log\left(\frac{1}{10}\right) \neq \frac{\log 1}{\log 10}$

$-1 \neq \frac{0}{1}$

$(\log_a x)^y \neq y\log_a x$

Example: $(\log 100)^3 \neq 3 \log 100$

$2^3 \neq 3(2)$

$a^x = y \qquad x = \frac{\log y}{\log a}$

$A = P\left(1+\frac{r}{n}\right)^{nt}$

$A$ = Final Amount after $t$ years
$r$ = Interest Rate per year
$P$ = Principal
$n$ = Number of Terms per year

$A(t) = A_0\left(1+\frac{r}{n}\right)^{nt} \xrightarrow{n \to \infty} A(t) = A_0 e^{rt}$

$A(t)$ = Final Amount after $t$ years
$A_0$ = Initial Amount
$r$ = Rate of Increase ($+r$) / Decrease ($-r$) per year

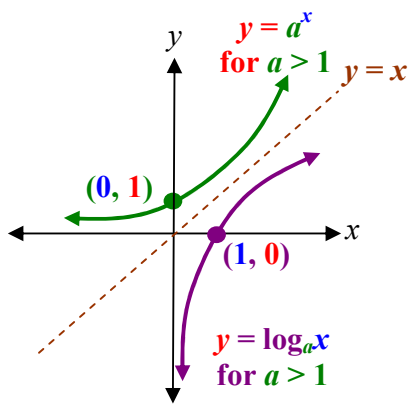
$A(t) = A_0 e^{rt}$

$A(t)$ = Final Amount after $t$ years
$A_0$ = Initial Amount
$r$ = Rate of Increase ($+r$) / Decrease ($-r$) per year

$N(t) = N_0 e^{rt}$

$N(t)$ = Final Population after $t$ years, hours, minutes, or seconds
$N_0$ = Initial Population
$r$ = Rate of Increase per year, hour, minute, or second

## Graphs of Exponential and Logarithmic Functions




**Exponential Function**

$y$-int = 1 No $x$-intercept

Domain $x \in R$ ; Range $y > 0$

**Logarithmic Function**

$x$-int = 1 No $y$-intercept

Domain $x > 0$ ; Range $y \in R$

To obtain equation for the inverse of an exponential function, we start with

$y = a^x$

$x = a^y$ (switch $x$ and $y$ for inverse)

$y = \log_a x$ (rearrange to solve for $y$)

$\pi$ rad = 180° OR $\frac{\pi}{180}$ rad = 1°

$y = a \sin k(x + b) + c$ $\qquad$ $y = a \cos k(x + b) + c$

$|a|$ = Amplitude $\qquad$ $c$ = Vertical Displacement (how far away from the $x$–axis)

$b$ = Horizontal Displacement (Phase Shift) $\qquad$ $b > 0$ (shifted left) $\qquad$ $b < 0$ (shifted right)

$k$ = number of complete cycles in $2\pi$ $\qquad$ Period = $\frac{2\pi}{k} = \frac{360°}{k}$

Range = Minimum $\leq y \leq$ Maximum

$y = a \sin [\omega (t + b)] + c$ $\qquad$ $y = a \cos [\omega (t + b)] + c$

$|a|$ = Amplitude $\qquad$ $c$ = Vertical Displacement (distance between *mid-line* and $t$-axis)

$b$ = Horizontal Displacement (Phase Shift) $\qquad$ $b > 0$ (shifted left) $\qquad$ $b < 0$ (shifted right)

$\omega$ = number of complete cycles in $2\pi$ $\qquad$ Period = $\frac{2\pi}{\omega}$ $\qquad$ Frequency = $\frac{\omega}{2\pi}$

Range = Minimum $\leq y \leq$ Maximum

Note: $\sin^{-1}(x) \neq \frac{1}{\sin(x)}$ $\qquad$ $\sin^{-1}(x) \neq (\sin x)^{-1}$ $\qquad$ $(\sin x)^{-1} = \frac{1}{\sin(x)} = \csc x$

$y = \sin^{-1} x$

Domain: $-1 \leq x \leq 1$ $\qquad$ Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$y = \cos^{-1} x$

Domain: $-1 \leq x \leq 1$ $\qquad$ Range: $0 \leq x \leq \pi$

$\sin(\sin^{-1} x) = x$ $\qquad$ for $-1 \leq x \leq 1$

$\sin^{-1}(\sin x) = x$ $\qquad$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$\cos(\cos^{-1} x) = x$ $\qquad$ for $-1 \leq x \leq 1$

$\cos^{-1}(\cos x) = x$ $\qquad$ for $0 \leq x \leq \pi$

$y = \tan^{-1} x$

Domain: $x \in R$ $\qquad$ Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$\tan(\tan^{-1} x) = x$ $\qquad$ for $x \in R$

$\tan^{-1}(\tan x) = x$ $\qquad$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

**Some Basic Trigonometric Definitions and Identities (proven equations)**

$\tan\theta = \frac{\sin\theta}{\cos\theta}$ $\qquad$ $\csc\theta = \frac{1}{\sin\theta}$ $\qquad$ $\sec\theta = \frac{1}{\cos\theta}$ $\qquad$ $\cot\theta = \frac{1}{\tan\theta} = \frac{\cos\theta}{\sin\theta}$

$\cos^2\theta + \sin^2\theta = 1$